

Periodic microlocal compactness forms retain the signed spectrum

Two counterexamples to the published Oscillation Lemma and an exact replacement

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Abstract

Rindler asks which formal microlocal compactness forms (MCFs) admit a generating sequence. We prove an exact formula for the basic realizable class generated by bounded one-periodic profiles. The formula is the signed Fourier cross-spectrum of $h \circ w$ and w , not the Young-measure expression stated in Lemma 4.1 of the source. Two counterexamples are decisive. A real two-phase square wave makes the published formula twice the defining MCF pairing. A positive-frequency complex exponential has zero negative-cone pairing although the published formula assigns it nonzero mass; hence no scalar normalization repairs the claim. The full generability problem is not claimed, but any characterization must retain this signed cross-spectral compatibility.

1 Source question and source claim

The source is F. Rindler, *Directional oscillations, concentrations, and compensated compactness via microlocal compactness forms*, arXiv:1212.1430v3, published in *Archive for Rational Mechanics and Analysis* 215 (2015), 1–63 [1].

After Example 5.8 (printed p. 38), the source observes that not every formal MCF is generated and asks for a characterization of the generated subclass:

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We end this section with an example showing that certain MCFs cannot be generated by *any* sequence. This raises the problem of characterizing the subclass of MCFs that are generated by at least one sequence. We reserve this for future work.

Example 5.8. Let

$$a := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad b := \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad A := a \otimes a, \quad B := b \otimes b,$$

and consider the MCF

$$\omega = \mathcal{L}^d \llcorner \Omega \otimes \left[\frac{1}{2} A \delta_A \otimes \bar{\delta}_{\pm a} + \frac{1}{2} B \delta_B \otimes \bar{\delta}_{\pm b} \right].$$

A quick way to see that this ω is not generated by any sequence is as follows: Theorem 5.3 implies that any generating sequence asymptotically would have to be a sequence of gradients. However, the corresponding Young measure is $\frac{1}{2} \delta_A + \frac{1}{2} \delta_B$, which is not a gradient Young measure. This follows for example from the Kinderlehrer–Pedregal theorem [KP91, KP94], because $\text{rank}(A - B) = 2$.

The basic asserted supply of generated forms is Lemma 4.1 (printed p. 28):

4.1. Oscillations. Consider the prototypical oscillating sequence

$$u_j(x) = A \mathbb{1}_{(0,\theta)}(jx \cdot n_0 - \lfloor jx \cdot n_0 \rfloor) + B \mathbb{1}_{(\theta,1)}(jx \cdot n_0 - \lfloor jx \cdot n_0 \rfloor), \quad x \in \Omega, \quad (4.1)$$

where $A, B \in \mathbb{C}^N$, $n_0 \in \mathbb{S}^{d-1}$, $\mathbb{1}_{(0,\theta)}$ is the indicator function of the interval $(0, \theta)$, and $\lfloor s \rfloor$ denotes the largest integer below $s \in \mathbb{R}$. Intuitively, this sequence oscillates between the values A and B , with volume fractions θ and $1 - \theta$, respectively, and in direction n_0 . The following general principle about oscillations shows that all of these facts can be recovered from the MCF generated by the sequence (u_j) :

Lemma 4.1 (Oscillations). *Let $w \in L^\infty(\mathbb{R}; \mathbb{C}^N)$ be 1-periodic (the “profile function”) and assume that $w_j(s) := w(js)$ ($s \in \mathbb{R}$) generates the homogeneous Young measure $\nu \in \mathbf{M}^1(\mathbb{C}^N)$. Then, the simple oscillation in direction $n_0 \in \mathbb{S}^{d-1}$,*

$$u_j(x) := w(jx \cdot n_0), \quad x \in \Omega, \quad (4.2)$$

generates the microlocal compactness form

$$\omega = \mathcal{L}^d \llcorner \Omega \otimes \left[\overline{(z - Z_0)} \nu(dz) \right] \otimes \bar{\delta}_{\pm n_0} \in \mathbf{MCF}^2(\Omega; \mathbb{C}^N),$$

where $Z_0 := \int_0^1 w ds$ is the average of w over one period cell and $\bar{\delta}_{\pm n_0} := \bar{\delta}_{-n_0} + \bar{\delta}_{+n_0}$. This notation means that for all $f \in \mathbf{F}^2(\Omega; \mathbb{C}^N)$, $\Psi \in \mathcal{M}$,

$$\langle\langle f \otimes \bar{\Psi}, \omega \rangle\rangle = \int_{\Omega} \int h(x, z) \cdot \overline{[\Psi(+n_0) + \Psi(-n_0)](z - Z_0)} \, d\nu(z) \, dx.$$

In the source notation, a bounded periodic profile w with mean Z_0 and Young measure ν is asserted to give, for $u_j(x) = w(jx \cdot n_0)$,

$$\langle\langle \phi h \otimes \bar{\Psi}, \omega \rangle\rangle = \int_{\Omega} \phi(x) \int h(z) \cdot \overline{[\Psi(n_0) + \Psi(-n_0)](z - Z_0)} \, d\nu(z) \, dx. \quad (1)$$

2 Idea of the proof

The multiplier Ψ sees the sign of every nonzero Fourier mode. If c_k is the k th coefficient of w and a_k that of $h \circ w$, then frequency separation pairs a_k only with c_k , in direction $\text{sgn}(k)n_0$. Formula (1) instead replaces each one-sided part by the complete defect $w - Z_0$; it therefore counts a real two-phase defect twice and creates a nonexistent negative-frequency defect for a positive-frequency complex wave. Trigonometric polynomials give the correct formula directly; L^2 approximation and Cauchy–Schwarz complete the proof for bounded profiles.

3 Exact periodic-profile formula

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For $w \in L^\infty(\mathbb{T}; \mathbb{C}^N)$ and continuous $h : \mathbb{C}^N \rightarrow \mathbb{C}^N$, define

$$c_k = \int_0^1 w(s) e^{-2\pi i k s} \, ds, \quad a_k = \int_0^1 h(w(s)) e^{-2\pi i k s} \, ds.$$

Theorem 1 (signed cross-spectrum formula). *Let $\Omega \subset \mathbb{R}^d$ be bounded, $n_0 \in \mathbb{S}^{d-1}$, and $u_j(x) = w(jx \cdot n_0)$. Let ω be the 2-MCF generated by a subsequence. For every $\phi \in C_c^\infty(\Omega)$, every continuous h (only its restriction to the bounded range of w matters), and every admissible matrix multiplier Ψ ,*

$$\langle\langle \phi h \otimes \bar{\Psi}, \omega \rangle\rangle = \left(\int_{\Omega} \phi(x) \, dx \right) \sum_{k \in \mathbb{Z} \setminus \{0\}} a_k \cdot \overline{\Psi(\text{sgn}(k)n_0) c_k}. \quad (2)$$

The series is absolutely convergent. In particular the right side is subsequence-independent.

Proof. First suppose that w and $H := h \circ w$ are trigonometric polynomials. Localize inside Ω as in the source's multiplier-localization lemma, and factor a dense collection of spatial tests as $\phi = \phi_1 \overline{\phi_2}$ with band-limited Schwartz factors. Then

$$\mathcal{F}[\phi_1 H(j \cdot n_0)](\xi) = \sum_k a_k \widehat{\phi_1}(\xi - jkn_0), \quad \mathcal{F}[\phi_2 w(j \cdot n_0)](\xi) = \sum_l c_l \widehat{\phi_2}(\xi - jl n_0).$$

For large j , distinct translated supports are disjoint, so only $k = l$ survives in the Parseval pairing. The high-frequency cutoff removes $k = l = 0$. On the translate indexed by $k \neq 0$, the multiplier converges to $\Psi(\text{sgn}(k)n_0)$. The remaining spatial integral is $\int \phi$, proving (2) for trigonometric polynomials.

For general bounded w , both w and H lie in $L^2(\mathbb{T})$. Approximate each by trigonometric polynomials. Uniform L^2 multiplier bounds and periodic averaging control the error in the defining pairing. On the coefficient side,

$$\sum_{k \neq 0} |a_k| |\Psi(\text{sgn}(k)n_0) c_k| \leq \|\Psi\|_\infty \| (a_k) \|_{\ell^2} \| (c_k) \|_{\ell^2},$$

so the series converges absolutely and the polynomial formulas pass to the limit. Density of the localized spatial tests gives all $\phi \in C_c^\infty(\Omega)$. $\square$

For $\Psi \equiv I$, Parseval reduces (2) to

$$\int_0^1 h(w(s)) \cdot \overline{w(s) - c_0} ds,$$

the expected Young-measure defect. It occurs once, not once in each cone.

4 Two counterexamples

Corollary 2 (real two-phase contradiction). *Lemma 4.1 of [1] is false even for $d = N = 1$, real-valued w , and the constant multiplier.*

Proof. Let $w = 1$ on $(0, \frac{1}{2})$ and $w = -1$ on $(\frac{1}{2}, 1)$, periodically; take $h(z) = z$ and $\Psi \equiv 1$. Its mean is zero and its Young measure is $\nu = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1}$. By the MCF definition (or (2)), the pairing is $\int_\Omega \phi$: the low-frequency part is compact and $u_j \rightharpoonup 0$, while $|u_j| = 1$. But (1) gives

$$\int_\Omega \phi \int z \overline{(1+1)z} d\nu(z) = 2 \int_\Omega \phi,$$

a contradiction for any nonzero nonnegative $\phi \in C_c^\infty(\Omega)$. $\square$

Corollary 3 (one-sided contradiction). *The failure is not repaired by inserting a factor $1/2$ in $\bar{\delta}_{\pm n_0}$.*

Proof. Take $d = N = 1$, $w(s) = e^{2\pi i s}$, $h(z) = z$, and choose the multiplier on $\mathbb{S}^0 = \{-1, +1\}$ by $\Psi(+1) = 0$, $\Psi(-1) = 1$. In (2), only $c_1 = a_1 = 1$ is nonzero, so the pairing vanishes. Equivalently, on $\Omega = (0, 1)$ the zero extension satisfies

$$\widehat{u_j}(\xi) = \widehat{\mathbf{1}_{(0,1)}}(\xi - j), \quad \int_{\xi < 0} |\widehat{u_j}(\xi)|^2 d\xi = \int_{\eta < -j} |\widehat{\mathbf{1}_{(0,1)}}(\eta)|^2 d\eta \longrightarrow 0.$$

Thus its negative-frequency MCF component is zero. Yet $Z_0 = 0$, the Young measure is Haar probability on the unit circle, and (1) predicts $\int_\Omega \phi$. The source formula loses the signed spectrum, not merely a numerical factor. $\square$

5 Implications, verification, and scope

The erroneous step is equation (4.4) in the source proof: after localization to one cone, it replaces the cone projection of $u_j - Z_0$ by the entire $u_j - Z_0$. The correct term is the corresponding one-sided Fourier part. For a real two-state profile and affine tests, the familiar product formula can be repaired by assigning half the variance to each sign. For general profiles there is no product of the Young measure with a fixed two-point direction measure; the cross-spectrum (2) is essential.

The full characterization of all generated MCFs remains open. The theorem does provide an exact generated subclass and a necessary lesson for any general answer: the recovered Young measure and an unsigned directional support do not determine the nonlinear directional correlations. Source examples and lamination formulas that invoke Lemma 4.1 require re-audit; no claim is made here about which later conclusions survive after correction.

The two counterexamples were checked both from the defining double limit and from (2); a finite Fourier verifier is included in the packet. Exact-phrase, title, periodic-profile, correction, erratum, and generability searches on 13 August 2026 found the source and author slides repeating the same claim, but no published correction or full characterization. Novelty is plausible, not certified. Human review should focus on the localization step in Theorem 1 and on downstream uses of Lemma 4.1.

References

- [1] F. Rindler, *Directional oscillations, concentrations, and compensated compactness via microlocal compactness forms*, Arch. Ration. Mech. Anal. 215 (2015), 1–63; arXiv:1212.1430v3.